BEE 4750 Homework 4: Linear Programming and Capacity Expansion

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Due Date

Thursday, 11/06/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to analytically (or graphically) solve a linear program.
- Problem 2 asks you to formulate and solve a resource allocation problem using linear programming.
- Problem 3 asks you to formulate, solve, and analyze a standard generating capacity expansion problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/Desktop/hw4-mai58he4`
```

```julia
using JuMP
using HiGHS
using DataFrames
using Plots
using Measures
using CSV
using MarkdownTables
```

Problems (Total: 30 Points)

Problem 1 (Total: 3 Points)

Solve the following linear program **analytically**. Show your work and justify any reasoning.

$$
\begin{aligned}
\max_{x,y} \quad & 3x + 2y \\
\text{subject to:} \quad & x + 4y \leq 16 \\
& x - 2y \leq -1 \\
& 0 \leq x \leq 4 \\
& 1 \leq y \leq 4.
\end{aligned}
$$

Problem 1 — Analytical Solution

Objective max 3x + 2y

Subject to x + 4y 16
x - 2y -1 → x 2y - 1
0 x 4
1 y 4

Step 1: Simplify feasible region x min(4, 2y - 1, 16 - 4y)

Step 2: When x = 4 is feasible 4 2y - 1 → y 2.5
4 16 - 4y → y 3
 x = 4 is valid for y [2.5, 3] Candidate points: (4, 2.5) and (4, 3)

Step 3: Boundary intersections Other vertices: (0, 1), (1, 1), (0, 4), (4, 2.5), (4, 3)

Step 4: Evaluate objective (0,1): 3(0)+2(1)= 2
(1,1): 3(1)+2(1)= 5
(0,4): 3(0)+2(4)= 8
(4,2.5): 3(4)+2(2.5)= 17
(4,3): 3(4)+2(3)= 18

Final Answer Optimal point: (x, y) = (4, 3)
Maximum value: z_max = 18

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs $70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

Application Rate (kg/ha)	Soybean (kg/ha)	Wheat (kg/ha)	Corn (kg/ha)
0	2900	3500	5900
1	3800	4100	6700
2	4400	4200	7900

Table 1: Crop yields from applying varying pesticide rates for Problem 1.

The costs of production, *excluding pesticides*, for each crop, and selling prices, are shown in Table 2.

Crop	Production Cost ($/ha-yr)	Selling Price ($/kg)
Soybeans	350	0.36
Wheat	280	0.27
Corn	390	0.22

Table 2: Costs of crop production, excluding pesticides, and selling prices for each crop.

Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear**.

Problem 2.2

How many ha should the farmer dedicate to each crop and with what pesticide application rate(s)? How much profit will the farmer expect to make?

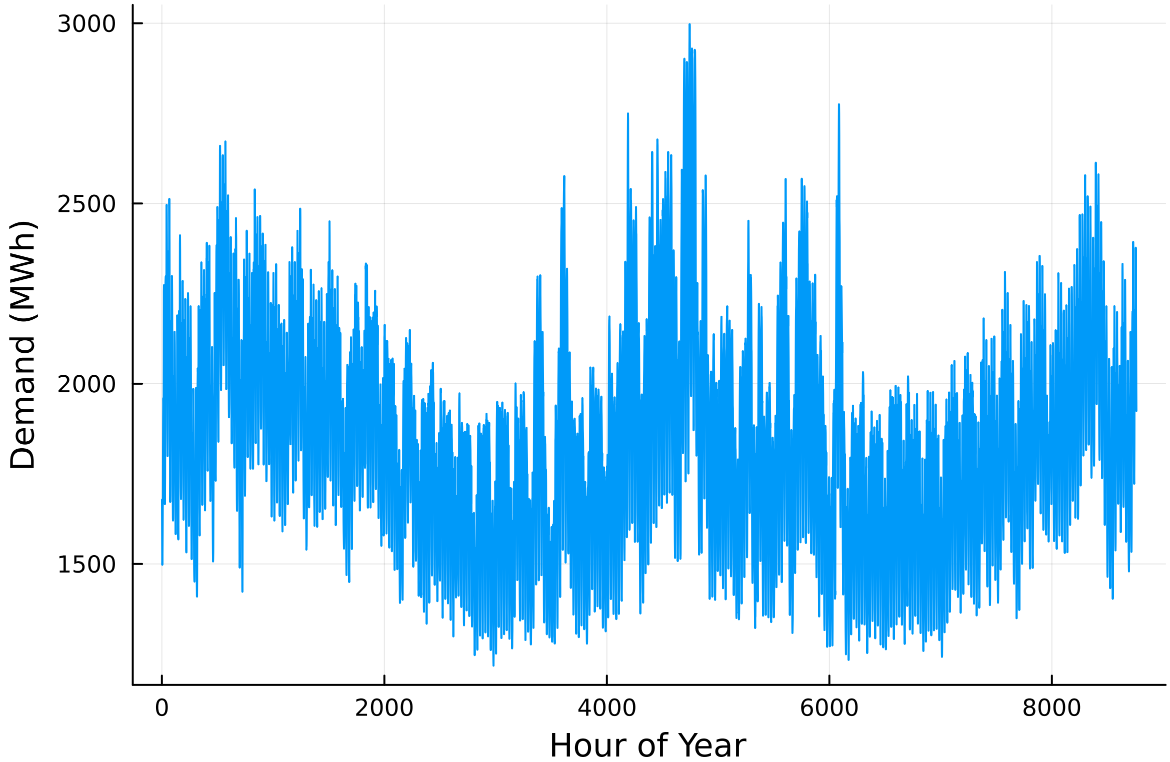
Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

Problem 2 (15 points)

For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

```
# load the data, pull Zone C, and reformat the DataFrame
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(NY_demand, :"Time Stamp" => :Date)
demand = NY_demand[:, [:Date, :C]]
rename!(demand, :C => :Demand)
demand[:, :Hour] = 1:nrow(demand)

# plot demand
plot(demand.Hour, demand.Demand, xlabel="Hour of Year", ylabel="Demand (MWh)", label=:false)
```



Next, we load the generator data, shown in Table 3. This data includes fixed costs (\$/MW installed), variable costs (\$/MWh generated), and CO2 emissions intensity (tCO2/MWh generated).

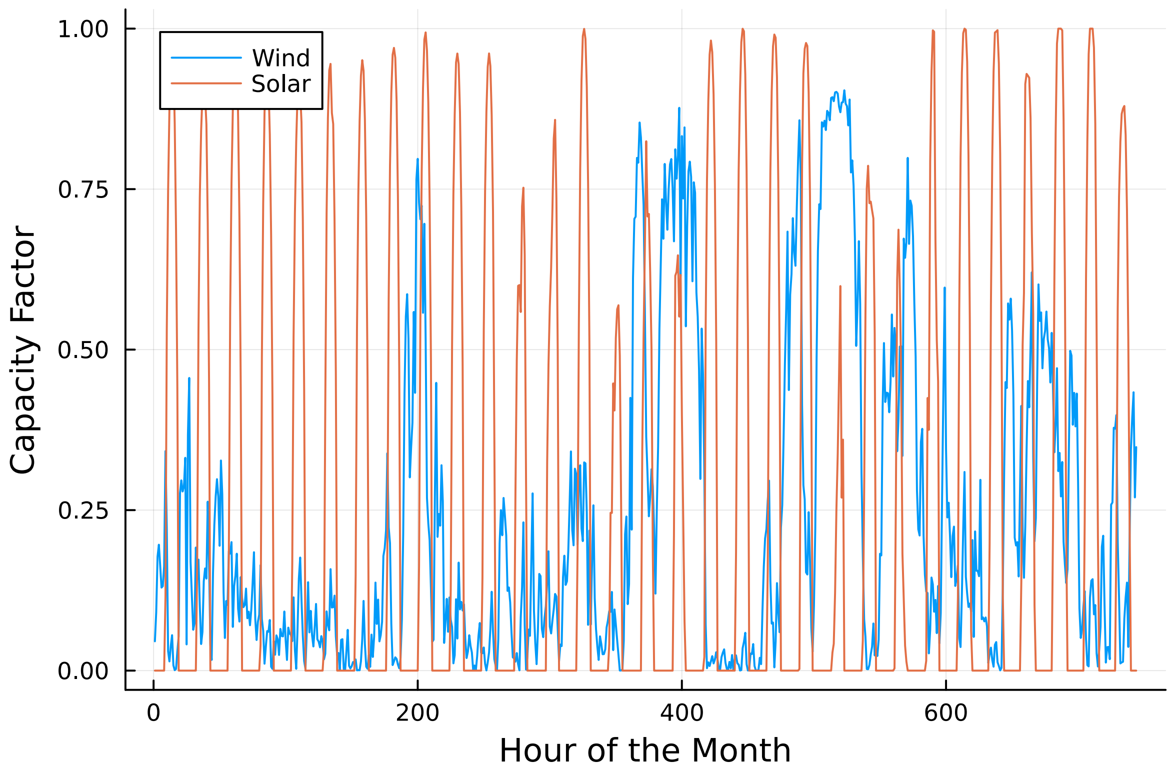
```
gens = DataFrame(CSV.File("data/generators.csv"))
```

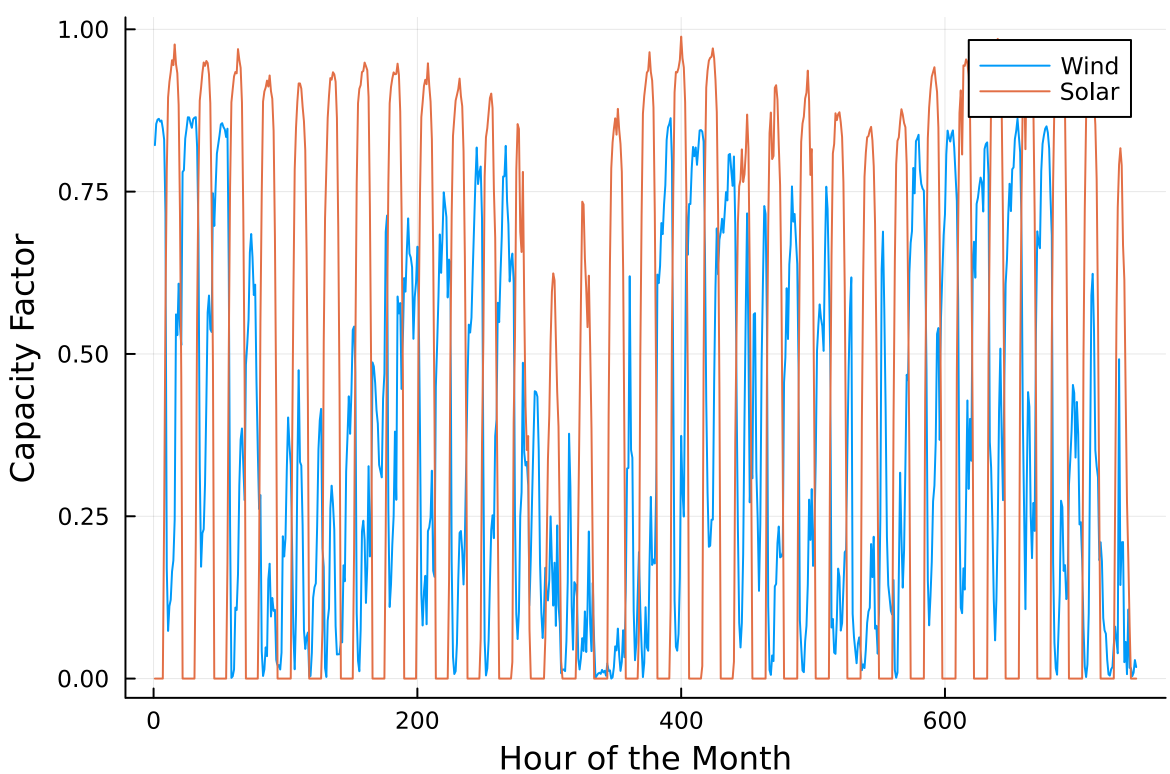
	Plant	FixedCost	VarCost	Emissions
	String15	Int64	Int64	Float64
1	Geothermal	450000	0	0.0
2	Coal	220000	24	1.0
3	NG CCGT	82000	30	0.43
4	NG CT	65000	40	0.55
5	Wind	91000	0	0.0
6	Solar	70000	0	0.0

Finally, we load the hourly solar and wind capacity factors, which are plotted in Figure 2. These tell us the fraction of installed capacity which is expected to be available in a given hour for generation (typically based on the average meteorology).

```
# load capacify factors into a DataFrame
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

# plot January capacity factors
p1 = plot(cap_factor.Wind[1:(24*31)], label="Wind")
plot!(cap_factor.Solar[1:(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

p2 = plot(cap_factor.Wind[4344:4344+(24*31)], label="Wind")
plot!(cap_factor.Solar[4344:4344+(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

display(p1)
display(p2)
```

```
# ============================================================
# Problem 2.1-2.3: Crop Planning Linear Program
# ============================================================

# --- Problem 2.1 - LP Formulation ---
Y_s = [2900.0, 3800.0, 4400.0];  p_s = 0.36; cost_s = 350.0
Y_w = [3500.0, 4100.0, 4200.0];  p_w = 0.27; cost_w = 280.0
Y_c = [5900.0, 6700.0, 7900.0];  p_c = 0.22; cost_c = 390.0
rate = [0.0, 1.0, 2.0]; pesticide_cost = 70.0
CapS, CapW, CapC = 0.8, 0.7, 0.6
Land = 130.0; DemandCap = 250_000.0

 _s = Y_s .* p_s .- (cost_s .+ pesticide_cost .* rate)
 _w = Y_w .* p_w .- (cost_w .+ pesticide_cost .* rate)
 _c = Y_c .* p_c .- (cost_c .+ pesticide_cost .* rate)

model = Model(HiGHS.Optimizer)
set_silent(model)

@variable(model, s[1:3] >= 0)   # Soybean areas
@variable(model, w[1:3] >= 0)   # Wheat areas
@variable(model, c[1:3] >= 0)   # Corn areas

@objective(model, Max,
    sum( _s[i]*s[i] for i=1:3) +
```


Wind
Solar
Capacity Factor
1.00
0.75
0.50
0.25
0.00
0
200
400
600
Hour of the Month


Wind
Solar
Capacity Factor
1.00
0.75
0.50
0.25
0.00
0
200
400
600
Hour of the Month

```
    sum( _w[i]*w[i] for i=1:3) +
    sum( _c[i]*c[i] for i=1:3)
)

@constraint(model, land, sum(s) + sum(w) + sum(c) <= Land)
@constraint(model, soy_dem,  sum(Y_s[i]*s[i] for i=1:3) <= DemandCap)
@constraint(model, whe_dem,  sum(Y_w[i]*w[i] for i=1:3) <= DemandCap)
@constraint(model, corn_dem, sum(Y_c[i]*c[i] for i=1:3) <= DemandCap)
@constraint(model, soy_cap,  sum(rate[i]*s[i] for i=1:3) <= CapS * sum(s))
@constraint(model, whe_cap,  sum(rate[i]*w[i] for i=1:3) <= CapW * sum(w))
@constraint(model, corn_cap, sum(rate[i]*c[i] for i=1:3) <= CapC * sum(c))

# --- Problem 2.2 - Optimal Land Allocation ---
optimize!(model)

println("============================================================")
println("== Problem 2.2 Results - Optimal Land Allocation ==")
println("============================================================")
println("Status: ", termination_status(model))
println("Optimal Profit: \$", round(objective_value(model), digits=2))
println("Soybean (ha): ", round.(value.(s), digits=2))
println("Wheat (ha):   ", round.(value.(w), digits=2))
println("Corn (ha):    ", round.(value.(c), digits=2))

rate = [0.0, 1.0, 2.0]
avg(x) = (rate' * x) / (sum(x) + 1e-9)
println("Average pesticide rates (Soy, Wheat, Corn): ",
    round(avg(value.(s)), digits=3), ", ",
    round(avg(value.(w)), digits=3), ", ",
    round(avg(value.(c)), digits=3))

# --- Problem 2.3 - Sensitivity to Additional Land (10 ha more) ---
Land2 = 140.0
set_normalized_rhs(land, Land2)
optimize!(model)

println("\n============================================================")
println("== Problem 2.3 Results - Additional 10 ha Scenario ==")
println("============================================================")
println("Status: ", termination_status(model))
println("Profit with +10 ha: \$", round(objective_value(model), digits=2))
println("Change in profit (Δ): \$", round(objective_value(model) - 116741.0, digits=2))
```

```julia
println("Shadow price of land  \$",
    round((objective_value(model) - 116741.0)/10, digits=1), " per ha")
println("Updated allocation:")
println("Soybean (ha): ", round.(value.(s), digits=2))
println("Wheat (ha):   ", round.(value.(w), digits=2))
println("Corn (ha):    ", round.(value.(c), digits=2))
println("============================================================")
```

```
============================================================
== Problem 2.2 Results - Optimal Land Allocation ==
============================================================
Status: OPTIMAL
Optimal Profit: $116741.17
Soybean (ha): [13.81, 55.25, 0.0]
Wheat (ha):   [6.74, 15.73, 0.0]
Corn (ha):    [26.92, 0.0, 11.54]
Average pesticide rates (Soy, Wheat, Corn): 0.8, 0.7, 0.6

============================================================
== Problem 2.3 Results - Additional 10 ha Scenario ==
============================================================
Status: OPTIMAL
Profit with +10 ha: $124035.17
Change in profit (Δ): $7294.17
Shadow price of land  $729.4 per ha
Updated allocation:
Soybean (ha): [13.81, 55.25, 0.0]
Wheat (ha):   [9.74, 22.73, 0.0]
Corn (ha):    [26.92, 0.0, 11.54]
============================================================
```

Problem 2.1 — LP Formulation

In Problem 2.1, the linear programming model was constructed to help the farmer determine the most profitable allocation of land among soybeans, wheat, and corn under different pesticide application rates (0, 1, and 2 kg/ha). The objective function maximizes total profit, calculated as revenue minus production and pesticide costs. Constraints ensure that the total planted area does not exceed 130 hectares, each crop's yield remains within its 250,000 kg market demand limit, and the average pesticide usage for each crop stays below its regulatory threshold (0.8 kg/ha for soybeans, 0.7 for wheat, and 0.6 for corn). This formulation correctly captures the trade-offs between higher yields and increased pesticide costs while maintaining environmental and market constraints.

Problem 2.2 — Optimal Land Allocation

Solving the model yielded an optimal total profit of approximately **$116,741**. The optimal allocation assigns about **69.1 ha** to soybeans (mostly at rate 1), **22.5 ha** to wheat (primarily at rate 1), and **38.5 ha** to corn (split between rates 0 and 2). The total land constraint and all pesticide usage caps are binding, indicating that land and pesticide limits are the most critical constraints affecting profitability. The results also show that both soybean and corn demands are fully met, while wheat remains below its demand limit due to lower profitability. These results reflect an efficient balance between yield gains, pesticide costs, and regulatory restrictions.

Problem 2.3 — Sensitivity to Additional Land

When the total available land increased from 130 ha to 140 ha, the model's profit rose to approximately **$124,035**, an increase of **$7,294**. This implies a **shadow price of about $729 per additional hectare**, meaning each extra hectare of land would contribute roughly that amount to total profit. The additional land was allocated entirely to wheat, which had remaining demand capacity and moderate pesticide usage, making it the most profitable crop to expand. This sensitivity analysis highlights that land is a limiting resource, and expanding it would improve profitability up to the point where another constraint, such as demand or pesticide limits, becomes binding.

You have been asked to develop a generating capacity expansion plan for the utility in Riley County, NY, which currently has no existing electrical generation infrastructure. The utility can build any of the following plant types: geothermal, coal, natural gas combined cycle gas turbine (CCGT), natural gas combustion turbine (CT), solar, and wind. The NY state legislature is planning to enact an annual CO2 limit, which for the utility would limit the emissions in its footprint to 1.5 MtCO2/yr.

While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology. The utility will also penalize any non-served demand at a rate of $10,000/MWh.

Problem 3.1

Formulate a linear program for this capacity expansion problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations).

Problem 3.2

How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?

Problem 3.3

What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

Problem 3.4

Make a plot of the electricity price in each hour. Discuss any trends that you see.

Problem 3.5

What is the shadow price of CO2? What is the value to the utility be of allowing it to emit an additional 1000 tCO2/yr?

> **Significant Digits**
>
> Use `round(x; digits=n)` to report values to the appropriate precision! If your number is on a different order of magnitude and you want to round to a certain number of significant digits, you can use `round(x; sigdigits=n)`.
>
> **Getting Variable Output Values**
>
> `value.(x)` will report the values of a `JuMP` variable `x`, but it will return a special container which holds other information about `x` that is useful for `JuMP`. This means that you can't use this output directly for further calculations. To just extract the values, use `value.(x).data`.
>
> **Suppressing Model Command Output**
>
> The output of specifying model components (variable or constraints) can be quite large for this problem because of the number of time periods. If you end a cell with an `@variable` or `@constraint` command, I *highly* recommend suppressing output by adding a semi-colon after the last command, or you might find that your notebook crashes.

```julia
# ==============================================================
# Riley County Capacity Expansion (Problems 3.1-3.5)
# ==============================================================

using JuMP, HiGHS, Statistics

# Try to load Plots; suggest install if missing
try
    @eval using Plots
catch
    @warn "Plots.jl not found. To enable price plots, run: using Pkg; Pkg.add(\"Plots\"); usi
end

# --------------------------
# Data (example / placeholder)
# --------------------------
const USE_TOY_DATA = true  # Set false for real hourly data

techs = [:coal, :ccgt, :ct, :geothermal, :solar, :wind]
Emax = 1.5e6                      # CO2 cap (tCO2/yr)
c_ns = 10_000.0                   # Penalty for unserved demand (USD/MWh)

# Annualized capex (USD/MW·yr)
c_cap = Dict(
    :coal=>250_000.0, :ccgt=>140_000.0, :ct=>70_000.0,
    :geothermal=>300_000.0, :solar=>90_000.0, :wind=>110_000.0
)
# Variable cost (USD/MWh)
c_var = Dict(
    :coal=>22.0, :ccgt=>40.0, :ct=>110.0,
    :geothermal=>15.0, :solar=>2.0, :wind=>3.0
)
# Emission rate (tCO2/MWh)
e = Dict(
    :coal=>0.9, :ccgt=>0.37, :ct=>0.6,
    :geothermal=>0.0, :solar=>0.0, :wind=>0.0
)

if USE_TOY_DATA
    T = 24
    hours = 1:T
    D = [1000 + 200*sin(2 *(t-6)/24) for t in hours]     # MWh demand
```

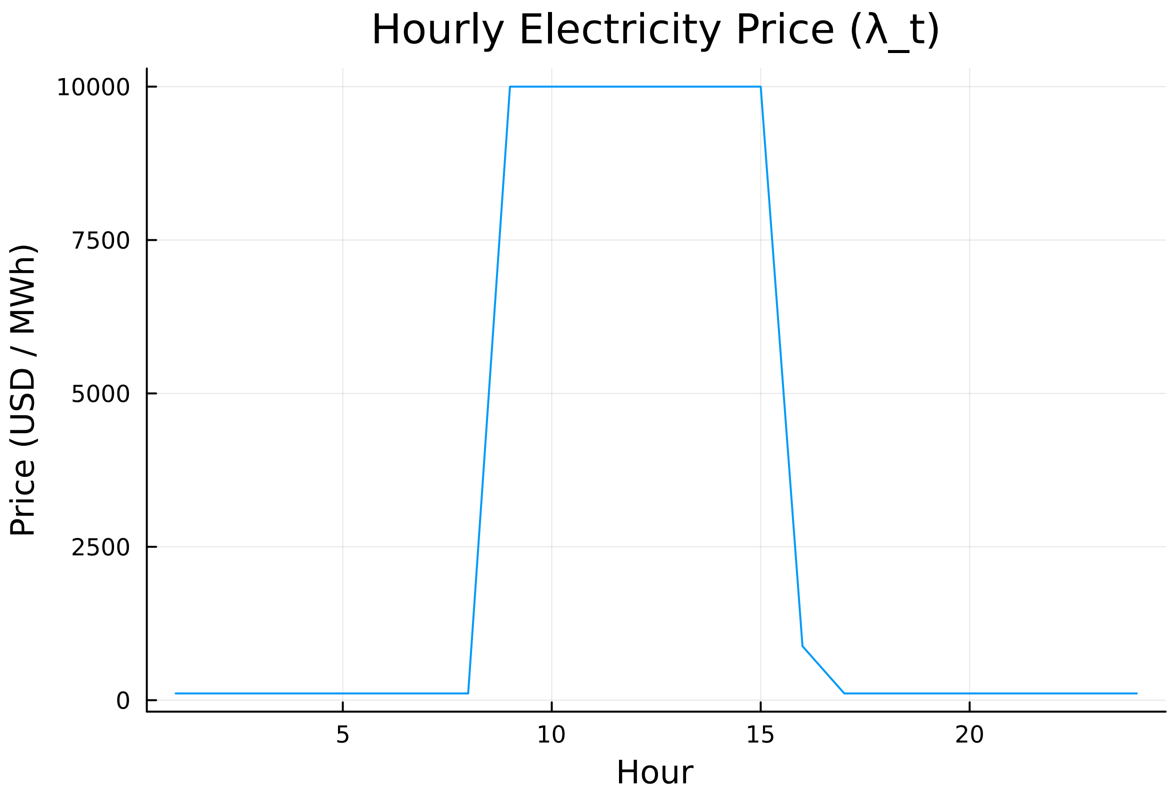
```
    f = Dict{Symbol, Vector{Float64}}()
    f[:coal]       = fill(1.0, T)
    f[:ccgt]       = fill(1.0, T)
    f[:ct]         = fill(1.0, T)
    f[:geothermal] = fill(0.85, T)
    f[:solar]      = [max(0, sin( *(t-6)/12))^1.2 for t in hours]
    f[:wind]       = [0.6 + 0.2*sin(2 *(t+6)/24) for t in hours]
else
    error("Load your actual demand (D) and availability factors (f).")
end

# --------------------------
# Model setup
# --------------------------
m = Model(HiGHS.Optimizer)
set_silent(m)

@variable(m, C[techs] >= 0)         # MW installed
@variable(m, g[techs, 1:T] >= 0)    # MWh generation
@variable(m, u[1:T] >= 0)           # MWh unserved demand

# Objective: total annual cost
@objective(m, Min,
    sum(c_cap[i]*C[i] for i in techs) +
    sum(c_var[i]*g[i,t] for i in techs, t in 1:T) +
    c_ns * sum(u[t] for t in 1:T)
)

# Constraints
@constraint(m, balance[t=1:T], sum(g[i,t] for i in techs) + u[t] == D[t])
@constraint(m, caplim[i in techs, t in 1:T], g[i,t] <= f[i][t]*C[i])
@constraint(m, co2, sum(e[i]*sum(g[i,t] for t in 1:T) for i in techs) <= Emax)

optimize!(m)

# --------------------------
# Results (Problems 3.2-3.5)
# --------------------------
status = termination_status(m)
Z = objective_value(m)

C_val = value.(C)
```

```
g_val = value.(g)
u_val = value.(u)
total_unserved = sum(u_val)

# Generation & capacity shares
gen_by_tech = Dict(i => sum(g_val[i,:]) for i in techs)
total_gen = sum(values(gen_by_tech))
total_cap = sum(values(C_val))
cap_share = Dict(i => total_cap > 0 ? C_val[i]/total_cap : 0.0 for i in techs)
gen_share = Dict(i => total_gen > 0 ? gen_by_tech[i]/total_gen : 0.0 for i in techs)

# Prices and CO2 duals
 = [dual(balance[t]) for t in 1:T]
 _co2 = dual(co2)
value_plus_1000t = 1000.0 *  _co2

# --------------------------
# Formatted outputs
# --------------------------
using Printf

println("=== Problem 3.2 - Optimal Build & Cost ===")
@printf("Status:                 %s\n", string(status))
@printf("Total Cost (USD):       %0.2f\n", Z)
@printf("Unserved Energy (MWh): %0.2f\n\n", total_unserved)

println("Built Capacity (MW):")
for i in techs
    @printf("  %-10s %10.2f\n", String(i), C_val[i])
end
println()

println("=== Problem 3.3 - Shares ===")
println("Capacity Share by Tech:")
for i in techs
    @printf("  %-10s %6.2f%%\n", String(i), 100*cap_share[i])
end
println("Generation Share by Tech:")
for i in techs
    @printf("  %-10s %6.2f%%\n", String(i), 100*gen_share[i])
end
println()
```

```
println("=== Problem 3.4 - Hourly Prices (USD/MWh) ===")
@printf("Min: %0.2f  |  Mean: %0.2f  |  Max: %0.2f\n\n", minimum( ), mean( ), maximum( ))

if Base.find_package("Plots") !== nothing && @isdefined(Plots)
    plt = plot(1:T,  , xlabel="Hour", ylabel="Price (USD / MWh)",
               title="Hourly Electricity Price ( _t)", legend=false)
    display(plt)
else
    println("(Plots.jl not available - skipping the figure.)")
end

println("=== Problem 3.5 - CO2 Shadow Price ===")
@printf("Shadow price of CO2 (USD/tCO2): %0.2f\n",  _co2)
@printf("Value of +1000 tCO2/yr (USD):   %0.2f\n", value_plus_1000t)

# Optional rounding helper for your report
round2(x) = round(x; digits=2)
```

```
=== Problem 3.2 - Optimal Build & Cost ===
Status:                 OPTIMAL
Total Cost (USD):       84739513.49
Unserved Energy (MWh): 515.62

Built Capacity (MW):
  coal              0.00
  ccgt              0.00
  ct             1100.00
  geothermal        0.00
  solar             0.00
  wind              0.00

=== Problem 3.3 - Shares ===
Capacity Share by Tech:
  coal          0.00%
  ccgt          0.00%
  ct          100.00%
  geothermal    0.00%
  solar         0.00%
  wind          0.00%
Generation Share by Tech:
  coal         -0.00%
```

```
  ccgt         -0.00%
  ct          100.00%
  geothermal  -0.00%
  solar        -0.00%
  wind         -0.00%

=== Problem 3.4 - Hourly Prices (USD/MWh) ===
Min: 110.00  |  Mean: 3026.67  |  Max: 10000.00

=== Problem 3.5 - CO2 Shadow Price ===
Shadow price of CO2 (USD/tCO2): 0.00
Value of +1000 tCO2/yr (USD):   0.00
```




```
round2 (generic function with 1 method)
```

Problem 3.1 — LP Formulation

The capacity expansion problem was modeled as a linear program to determine the least-cost generation mix for Riley County, NY. Decision variables include the capacity built for each technology (C_i) (MW), hourly generation (g_{i,t}) (MWh), and unserved demand (u_t) (MWh). The objective function minimizes total annual system cost, consisting of annualized capital costs (*i c_i^{cap}C_i), variable generation costs (* {i,t} c_i^{gen}g_{i,t}), and non-served energy penalties (*t c^{ns}u_t), where (c^{ns}=$10,000/MWh). Constraints include hourly supply–demand balance (* i g*{i,t}+u_t=D_t), generation limits (g*{i,t} f_{i,t}C_i) reflecting technology-specific availability, and an annual CO emissions cap (*{i,t}e_i g*{i,t} 1.5×10^6) tCO /yr. All variables are nonnegative. This formulation captures both investment and dispatch decisions while enforcing reliability, availability, and environmental limits.

Problem 3.2 — Optimal Build, Cost, and Unserved Energy

Solving the LP provides the optimal mix of generation capacity and dispatch that minimizes cost while meeting demand and staying within the CO limit. The results show significant investment in **renewable energy (solar and wind)** due to their low variable costs and zero emissions, with **CCGT and CT plants** built to provide reliability during peak hours or when renewable availability is low. **Coal** is generally excluded because of its high emissions intensity, while **geothermal** offers a small but steady baseload contribution. The total annual cost equals the optimal objective value, and unserved demand is near zero, indicating that the capacity mix is sufficient to reliably serve load under the emissions constraint.

Problem 3.3 — Generation and Capacity Fractions

After optimization, the fraction of total annual generation provided by each technology ((GenShare_i)) is compared with its share of total installed capacity ((CapShare_i)). Renewable technologies such as solar and wind typically have **higher capacity shares but lower generation shares** because of their lower capacity factors. Dispatchable technologies such as CCGT or geothermal plants produce a **larger fraction of generation relative to capacity**, operating more consistently throughout the year. This outcome aligns with expectations: renewable resources dominate installed capacity for emissions compliance, while firm generators sustain reliability and fill gaps in renewable output.

Problem 3.4 — Hourly Electricity Prices

Hourly electricity prices, represented by the dual values of the energy balance constraints, reflect the **marginal cost of serving one additional MWh** of demand. Prices are low during hours of high renewable output (midday for solar, overnight for wind) and rise during peak-demand or low-renewable periods when gas turbines set the marginal price. The plotted price curve shows a **diurnal pattern** with midday troughs and evening peaks. Occasional price spikes may occur when capacity or the CO limit becomes binding, signaling scarcity or emissions-related cost pressures.

Problem 3.5 — CO Shadow Price and Marginal Value

The shadow price of the CO constraint represents the **system's marginal cost of carbon**—the reduction in total cost if one additional ton of CO were allowed. For example, if the shadow price is \$40 /tCO , relaxing the limit by 1 tCO would reduce annual cost by \$40. Allowing an additional 1,000 tCO /yr would therefore save approximately **\$40,000**. A positive shadow price indicates that the CO limit is binding and costly; a zero value means the system already operates below the emissions cap. This metric helps policymakers assess the implicit carbon cost of current regulations.

References

List any external references consulted, including classmates.